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Research in Sports Medicine
An International Journal

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/gspm20

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To cite this article: Alexa E. Walter, James R. Wilkes, Madeleine Scaramuzzo, Tesa Johns-Bostick, Scott Lynch, Wayne Sebastianelli, Peter Seidenberg, Tim Bream & Semyon M. Slobounov (2022): Head acceleration event exposure and cognitive and functional outcomes: a comparison of multiple football seasons, Research in Sports Medicine, DOI: 10.1080/15438627.2022.2090249

To link to this article: https://doi.org/10.1080/15438627.2022.2090249

Published online: 16 Jun 2022.

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RESEARCH IN SPORTS MEDICINE
https://doi.org/10.1080/15438627.2022.2090249

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Head acceleration event exposure and cognitive and functional outcomes: a comparison of multiple football seasons

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ABSTRACT

Athletes in contact sports are exposed to repetitive impacts as an inherent part of sport. There is concern over the accumulative effect; however, much is still unknown regarding their short-term effects. This study investigated impact accumulation and outcomes over three seasons (2015, 2017, 2019) in NCAA Football Bowl Subdivision players. Impacts were recorded using helmet accelerometers, and virtual reality testing (VR) was done across the season. Incidence rates for impacts (total; ≥25 G to <80 G; ≥80 G) all significantly differed by season ($p < 0.05$). VR scores changed across the seasons, specifically significant decreases in spatial memory ($p < 0.05$) in 2015, significant changes in balance and spatial memory ($p < 0.05$) in 2017, and no significant changes in 2019. Linear regressions predicting VR change score by impact incidence rate were nonsignificant. Monitoring exposure to impacts and changes in outcomes is useful; however, results are fluid, and many factors could indirectly have protective effects on athletes.

ARTICLE HISTORY
Received 24 November 2021
Accepted 11 June 2022

KEYWORDS
Head acceleration events; subconcussive; football; outcomes; college-athlete

Introduction

Repetitive head impacts, or head acceleration events (HAE), are an inherent part of contact sports. They are defined as impacts that involve the transfer of mechanical energy to the brain with sufficient force to injure axonal or neuronal integrity but do not result in clinical symptoms (Bailes et al., 2013; Manley et al., 2017). Exposure to HAEs over a lifetime has been associated with later-in-life neurological dysfunction (McKee et al., 2016, 2009); however, presently, there is still much unknown about the short-term effects of exposure to HAEs.

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Individuals participating in football at the college level are at high risk for concussion (McCrea et al., 2021) and HAEs, with studies showing averages of 1000 impacts during a single season (Bailes et al., 2013). Yet studies examining the effect of HAE exposure on cognitive and functional outcomes have revealed mixed outcomes. In college athletes, few studies have demonstrated worsening functioning after a season of participation and the main domain often affected is balance (Kaminski et al., 2007; Miyashita et al., 2017). However, many studies have shown either no change (Caccese et al., 2019; Gysland et al., 2012; Murray et al., 2018) or improvements in score across seasons, especially on cognitive tasks (McAllister et al., 2012; Miller et al., 2007; Volberding & Melvin, 2014). Thus, it is still unknown whether changes reported in brain functioning are due to HAE exposure or other factors like fatigue, fitness level, or high-intensity training regimens over the season.

One potential confounder to these findings is that governing bodies, coaching, and medical staff can implement rule changes or safety procedures, assumed to protect athletes from injury and exposure to HAEs, during a season of play. Studies examining some of these rule changes have found both beneficial (kick-off rule change and decreased concussion rates (Wiebe et al., 2018)) and non-beneficial (elimination of two-a-day practices increased HAE exposure during preseason (Stemper et al., 2019)) effects on athlete health.

Given the mixed findings on exposure to HAEs and short-term functional and cognitive outcomes, this study aimed at examining the effect of HAEs on brain functionality after participation in a single season of football. By examining impact exposure and cognitive and functional domains both within a single season and over three non-consecutive seasons, we can study a) HAE exposure, b) functional and cognitive changes over a single season, c) how impact accumulation influences changes in outcome, and d) how, indirectly, changes made by medical and coaching personnel at a single institution, as well as the governing bodies of sport, could influence these outcomes of interest.

Methods

Participants

All participants were Penn State University NCAA Football Bowl Subdivision athletes, and background information is presented in Table 1. All participants provided informed written consent, as approved by the Pennsylvania State University Institutional Review Board.

Study procedures

This study was run in the non-consecutive competitive seasons of 2015, 2017, and 2019 and an overview of study procedures is presented in Figure 1.

HAE monitoring

Impacts to the head were monitored using helmet sensors from the Head Health Network BodiTrak system. These sensors were individually placed inside each individual participant's helmet using a 3M VHB adhesive and are comprised of elastic fabric with pressure monitors and impact sensors that output estimates of linear acceleration (in units of G). All

Table 1. Participant demographic information.

		2015	2017	2019	
N		23	34	31	
Age, *mean* ± *sd*		20.78 ± 1.31	20.49 ± 1.60	20.19 ± 1.28	anova $p = 0.323$
Years Played Football,*mean* ± *sd*		11.10 ± 3.46	11.21 ± 2.97	11.87 ± 3.16	anova $p = 0.670$
		N (%)	N (%)	N (%)	Chi Square
Position Category[a]	Non-speed	16 (69.6)	34 (100)	17 (54.8)	$p < 0.001$
	Speed	7 (30.4)	0 (0)	14 (45.2)	
History of Previous Concussion	No	14 (60.9)	18 (52.9)	19 (61.3)	$p = 0.751$
	Yes	9 (39.1)	16 (47.1)	12 (38.7)	
Number of previous concussions	0	14 (60.9)	18 (52.9)	19 (61.3)	$p = 0.855$
	1	7 (30.4)	13 (38.2)	9 (29.0)	
	2	2 (8.7)	2 (5.9)	3 (9.7)	
	3	0 (0)	1 (2.9)	0 (0)	
Learning Disability Diagnosis	No	20 (87.0)	23 (67.6)	24 (77.4)	$p = 0.142$
	ADD/ADHD	2 (8.7)	11 (32.4)	7 (22.6)	
	Dyslexia	1 (4.3)	0 (0)	0 (0)	
Psychiatric Disorder Diagnosis	No	22 (95.7)	33 (97.1)	29 (93.5)	$p = 0.438$
	Yes	0 (0)	1 (2.9)	2 (6.5)	

[a]Position categories based off of Lehman (2013). Speed positions include quarterback, running back, halfback, fullback, wide receiver, tight end, defensive back, safety, and linebacker, and non-speed positions include offensive and defensive linemen. Abbreviations: SD, standard deviation

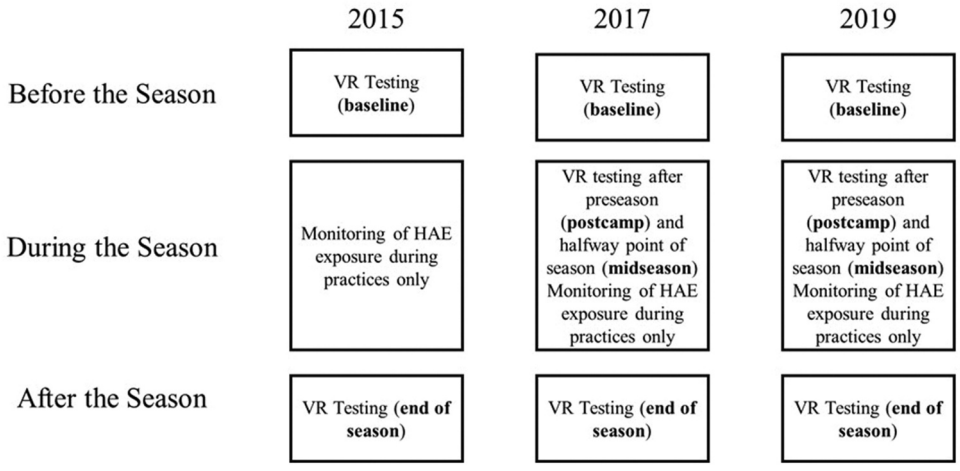

Figure 1. Schematic of study procedures for each competitive season.

sensors were placed by a certified athletic trainer who was trained on proper installation techniques and the sensors were monitored throughout the season for integrity and functionality. Impacts were categorized into total number of impacts, impacts ≥80 G, and impacts ≥25 G to <80 G based on work done by Mihalik et al. (2007) and on pilot work.

Virtual reality cognitive and functional outcome assessment

A 3D system (HeadRehab.com) with a head-mounted accelerometer was used. Three different modules were used: balance, reaction time, and spatial memory (see, Teel, Gay, Arnett et al., 2016; Teel, Gay, Johnson et al., 2016; Teel & Slobounov, 2015 for reliability and validity). In the balance task, participants hold tandem Romberg position

for all trials. In the first trial, the virtual room is completely still and in the subsequent 6 trials, the virtual room moves and the system measures position and orientation in yaw, pitch, and roll directions. In the reaction time module, participants stand feet shoulder width apart with hands on hips and are instructed to move their body in the same direction as the virtual room. The system measures reaction time (in ms) and errors of anticipation (wrong direction of response). In the spatial memory module, participants are shown a randomized virtual pathway to a door followed by the return trip. Afterwards, they are instructed to repeat the exact pathway using a joystick. The system measures errors to get to the door and total time to task completion. Raw data is used to produce a score on a scale of 0 (fail) to 10 (perfect).

Statistical analysis

All statistical analyses were performed using SPSS V25. For all analyses, statistical significance was considered reached at $p < 0.05$. To account for the different number of sessions between seasons, the incidence rate of impacts (total; ≥80 G; ≥25 G and <80 G) was calculated (number of impacts/(population size*number of sessions)) and used in analyses. Welch's ANOVA with Games–Howell post-hoc test was run to compare the incidence rates by season of play. VR results were assessed using a Wilcoxon Signed Rank test (for baseline to post-season changes) or a Friedman's test with post hoc analysis done using Wilcoxon Signed Rank tests with a Bonferroni correction (across season changes). Differences between seasons were examined using the Kruskal–Wallis test. Linear regressions were run to predict change score in VR modality (post-season – baseline) from incidence rate of total impacts.

Results

Accelerometer data

2015–129 impacts at ≥80 G were recorded across the season (53 sessions) with 78.3% occurring during the preseason (13 sessions) and 21.7% occurring during the regular season. A total of 3557 impacts at ≥25 G to <80 G were recorded across the season with 32.6% occurring during the preseason and 67.4% occurring during the regular season. The average incidence rate for all impacts was 0.1333 ± 0.0921, for impacts at ≥80 G was 0.0047 ± 0.0038, and for impacts ≥25 G to <80 G was 0.1228 ± 0.0878 (Figure 2).

2017–147 impacts at ≥80 G were recorded across the season (69 sessions) with 29.3% occurring during the preseason (13 sessions) and 70.7% occurring during the regular season. A total of 5733 impacts at ≥25 G to <80 G were recorded across the season with 27.8% occurring during the preseason and 72.2% occurring during the regular season. The average incidence rate for all impacts was 0.0737 ± 0.0379, for impacts ≥80 G was 0.0018 ± 0.0018, and for impacts ≥25 G to <80 G was 0.0719 ± 0.0375 (Figure 2).

2019–51 impacts at ≥80 G were recorded across the season (34 sessions) with 52.9% occurring during the preseason (14 sessions) and 47.1% occurring during the regular season. A total of 1332 impacts at ≥25 G to <80 G were recorded across the season with

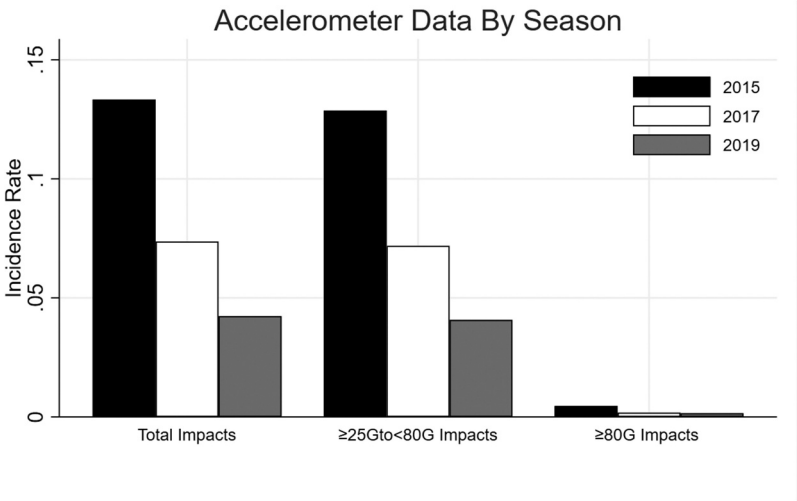

Figure 2. Incidence rate of impacts (total impacts; impacts ≥25 G and <80 G; impacts ≥80 G) over the seasons.

45.6% occurring during the preseason and 54.4% occurring during the regular season. The average incidence rate for all impacts was 0.0423 ± 0.0414, for impacts at ≥80 G was 0.0015 ± 0.0022, and for impacts ≥25 G to <80 G was 0.0408 ± 0.0398 (Figure 2).

Comparison between seasons – There was a significant difference in total impact incidence rate by season ($F = 11.71$, $p < 0.001$) and ≥25 G and <80 G incidence rate ($F = 11.68$, $p < 0.001$), with post-hoc testing revealing significant decreases between each of the seasons ($p < 0.05$). For ≥80 G incidence rate, there was also a significant difference by season ($F = 6.01$, $p < 0.01$), with differences only between 2015 and 2017 ($p = 0.008$), and 2015 and 2019 ($p = 0.005$).

Virtual reality data

VR results were produced on a scale of 0 (fail) to 10 (perfect) for each module: spatial memory, balance, and reaction time (Figure 3).

2015 – Data was examined at baseline and end of season and there were non-significant changes from baseline to end of season in balance and reaction time ($p > 0.05$) and a significant decrease ($Z = -2.160$, $p = 0.03$) in the spatial memory module (Walter et al., 2020).

2017 – There was a significant difference in balance score across the season $\chi^2(3) = 8.131$, $p = 0.043$. Pairwise comparisons revealed a significant decrease between the postcamp and end of season balance scores ($p = 0.034$). The spatial memory module showed a statistically significant change across the season $\chi^2(3) = 9.821$, $p = 0.020$. Pairwise comparisons revealed significant increase between baseline and midseason ($p = 0.034$). Additionally, there was a nonsignificant difference in reaction time score across the season ($p = 0.095$).

2019 – There were nonsignificant differences across the season in balance scores ($p = 0.250$), in spatial memory scores ($p = 0.427$), and in reaction time scores ($p = 0.932$).

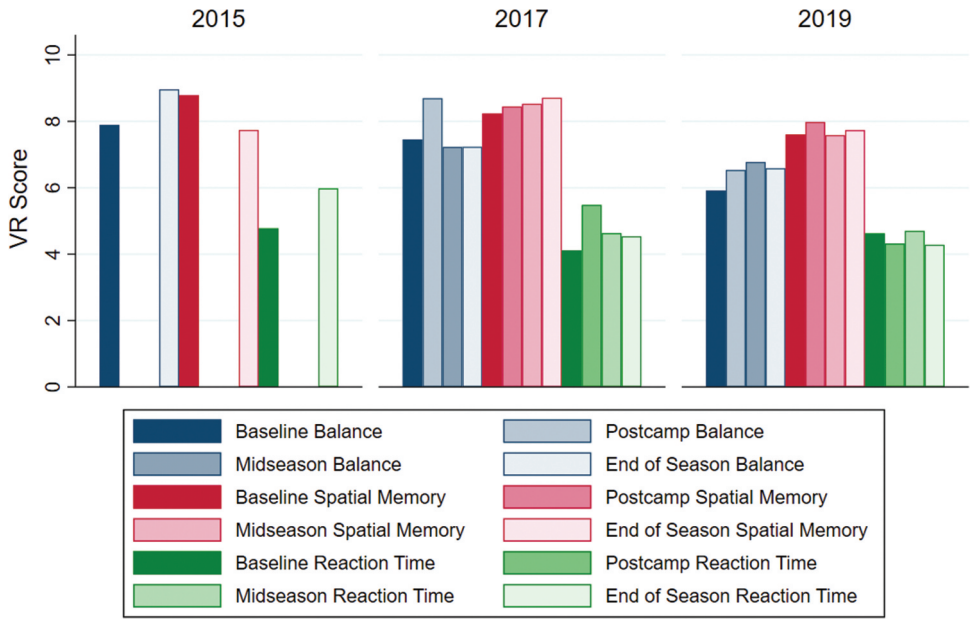

Figure 3. VR scores for balance, spatial memory, and reaction time modalities across time for each season.

Comparison between seasons – All comparisons for reaction time and spatial memory between the seasons were nonsignificant. For balance, there were significant differences at baseline (H = 10.514, p = 0.005), specifically between 2015 and 2019 (p = 0.008), and 2017 and 2019 (p = 0.039); at postcamp (H = 9.407, p = 0.002) between 2017 and 2019; and at end of season (H = 6.630, p = 0.036); however, all pairwise comparisons were nonsignificant.

Virtual reality change scores and total impacts

Linear regressions predicting VR change score (post-season – baseline) by impact incidence rates were run for the whole cohort (n = 87). None of the VR modalities were significantly predicated by total impact incidence rate.

Discussion

This study adds to the growing literature examining cognitive and functional changes over a single football season. By implementing testing across a season, we can better examine exposure to HAEs and how that may relate to changes in outcome. Additionally, over the course of three seasons, this paper allowed for a preliminary examination of reproducibility and how, indirectly, changes made at coaching, medical, or governing body levels can influence outcomes.

When using college athletes as participants in research, certain factors are out of researchers' control including medical and coaching decisions made by the individual team and rule changes made by governing bodies, like the National Collegiate Athletic

Association (NCAA). During the course of these seasons, there were important changes that need to be discussed and considered when interpreting these data. The NCAA implemented two rules changes that could influence these data: instant replay for targeting and the limiting of multiple contact practices a day during preseason. Additionally, decisions made at the Penn State level that could influence the way these data are interpreted were changes to practice format (the number of hitting practices a week), changes to coaching staff (new position coaches), and new equipment (helmets specifically chosen for each individual based on fit and the use of protective equipment including the Protech helmet cover). While these factors will most likely directly influence data, there is no way to quantity their individual or additive effects on these data. As such, they should be considered indirect effects on data and as part of a larger scope discussion.

Overall, raw numbers of impacts decreased across seasons. Additionally, the proportion of impacts during the preseason was substantially higher in 2015 (78.3%) compared to 2017 (29.3%) and 2019 (52.9%) suggesting that rule changes, coaching changes, or wearing a protective Protech helmet cover may have been beneficial at a single institution, as opposed to findings from other studies (Stemper et al., 2019).

However, as there were different numbers of sessions and players, incidence rate was used to better directly compare the data across seasons. There was a significant decrease in the incidence rate for all impacts and lower threshold impacts (≥25 G&<80 G) across time. The higher intensity impacts (≥80 G) decreased significantly from 2015 to 2017, and to 2019 but not from 2017 to 2019. These changes across seasons could be due to many factors such as position, or the way drills are structured; however, there is no way to directly quantify these effects.

By season, VR data from 2015 revealed a significant decrease in spatial memory scores from baseline to end of the season as well as non-significant increases in balance and reaction time scores. The decline in cognitive function is in line with previous reports that concussion may result in lower cognitive functions, including memory changes (Rabadi & Jordan, 2001) and adds to the growing body of literature examining exposure of HAE on cognitive functioning.

As baseline to post-season changes may not be the most sensitive indicator of the effects of HAEs, additional in season testing was added to the 2017 and 2019 protocols. In 2017, balance and spatial memory, but not reaction time, significantly changed over the season with balance significantly decreasing from post-camp to the end of the season and spatial memory significantly increasing from baseline to midseason. In 2019, there were no significant changes in any modality over the season. These findings reflect the complicated nature of using a single modality to study the influence of HAEs. There were no consistent findings across the three seasons and that could reflect a multitude of factors (less HAE exposure, different participant populations, fatigue/stress, lifting schedules, etc.). Many of these factors have the ability to affect testing results, so more frequent monitoring could be beneficial. Balance, in particular, is a more functional task, and while likely to be affected by physical fatigue, has been shown to be negatively affected by a season of exposure to HAEs (Kaminski et al., 2007; Miyashita et al., 2017; Munce et al., 2014).

When directly comparing the data between seasons, some interesting findings emerge. Balance was the only modality to significantly differ between seasons at baseline (2019 performed worse than both 2015 and 2017) and at post-camp (2019 performed

worse than 2017). These differences could be multifactorial ranging from player selection and sample size differences to indirect influences like practice schedules, strength and conditioning load, and coaching personnel changes. However, this suggests another interesting line of research to continue as balance reflects a more functional outcome as opposed to a purely cognitive outcome and as mentioned, multifactorial testing with both components may prove more beneficial.

Change scores in VR were not predicted by the total impact incidence rate for all players under study. This could be due to player selection or subtle, unmeasurable differences between the seasons, but it does corroborate findings from a study on multiple seasons of play in children participating in football (Rose et al., 2021).

Limitations

The present studies were based on small cohorts of football players; therefore, future work at continuing to replicate these findings as well as examining a greater variety of players in other sports (including females) would prove beneficial. Furthermore, only practice accelerometer data, and not game data, were collected which could confound the results. Future studies should examine these sessions as well, as there is potential for higher overall number of impacts, especially those of higher intensities. Lastly, there is concern over the use of accelerometers in contact sports due to the accuracy of the systems used. Forces measured by the helmet systems may not accurately represent the forces experienced by the brain itself. The system used in this study likely has inconsistencies in forces measured and margins of error like many other systems used in research.

Conclusions

While the data presented here are not intended to be used in a diagnostic capacity, it does have potential implications in decision-making on how to best protect the health of athletes. This study highlights the potential utility of a more wide-range and inclusive battery of tests when examining athlete functioning. The use of cognitive and functional components when assessing athletes may have the ability to identify more nuanced changes that are occurring compared to the use of one single clinical test. In comparing data between seasons, it demonstrates the effect that, in this case, an individual school and its personnel can have on exposure to HAEs and the complexity of factors that could influence exposure. As coaches adapt new techniques and technologies and as rules continue to change, understanding the implications of these changes, whether positive or negative, is critical to providing medical care and ensuring the safety of athletes participating in contact and collision sports.

Disclosure statement

Author PS has received royalties for his contributions to textbooks. Author TB is currently employed by SAFR Sport Technologies who makes the Protech helmet cover. All remaining authors declare that they have no conflict of interest.

Funding

Funding
This study was partially supported by the National Collegiate Athletic Association Grant #106031.

Availability of data and material

Data are available upon reasonable request to the senior author, Dr Slobounov.

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